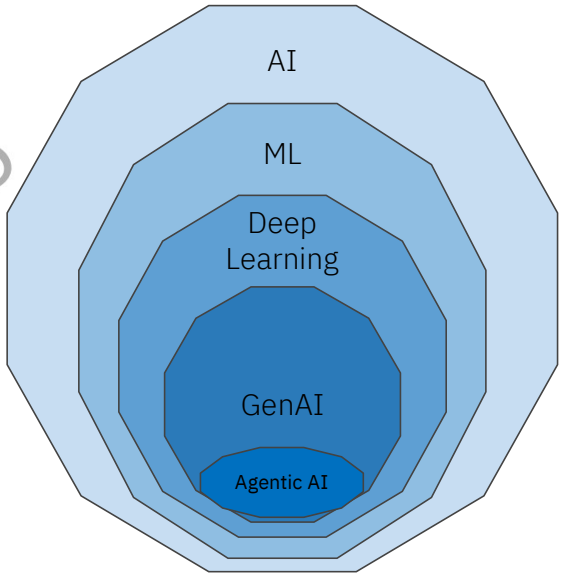
Introduction to AI

Week 2: Knowledge Representation for AI Learning

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090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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Objectives

- Describe how knowledge and intelligence are represented from data using models, rules, and structures such as neural networks or knowledge graphs.
- Explain how AI learns from data through processes like training, pattern recognition, and feedback.
- Identify ways to learn AI, including studying foundational concepts (data, algorithms, and logic), exploring AI tools and programming, and applying AI to real-world problems.

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AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation
W3: Search strategies

Machine Learning

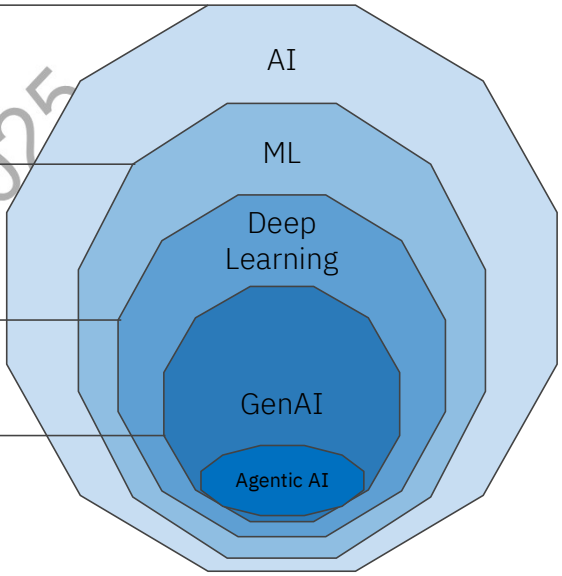
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



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Contents

- How AI learn?
- Knowledge Representation
- Reasoning
- How to learn AI?

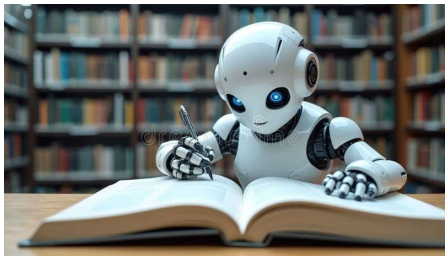
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How AI learn?

- How AI advise if we should go to play pickaball today?
- How AI classify emails as spam?
- How AI recognize a cat?
- How AI read your bike's plate?
- How AI guess your age?
- How AI predict your feelings?
- How AI classify your posts on social network?



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Learning

Humans	AI
• Learn by experience • Neuron + Memory • The process: experience -> knowledge -> intelligence	• Learn from (training) data • Model + Knowledge • The process: Data → Model → Prediction (→ Feedback → Improvement)

Knowledge Representation is how
AI organizes what it learns.

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What AI Can Do Today

Strengths

Pattern recognition: Detect faces in photos (Facebook, iPhone).

Speech recognition: Siri, Alexa, Google Assistant.

Game playing: AlphaGo beating world champions.

Recommendation systems: Netflix, Spotify, TikTok personalization.

Content generation: ChatGPT, DALL·E, Stable Diffusion (text, code, images).

Medical imaging: Detecting cancer in radiology scans with high accuracy.

Limitations

Common sense reasoning: LLMs can generate fluent answers but often hallucinate facts.

Generalization to new tasks: A model trained for chess can't suddenly play Go.

Causality & understanding: Correlates data patterns but doesn't truly "understand" why things happen.

Ethical failures: Bias in hiring tools, racial/gender bias in facial recognition.

Physical adaptability: Robots struggle in unstructured, unpredictable environments compared to humans.

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AI learns from experience

Supervised Learning (Learning from a "Teacher")

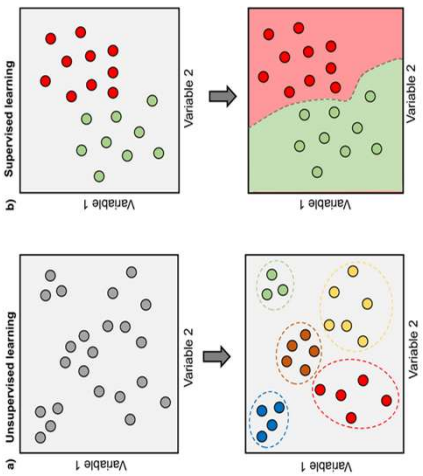
- *Goal*: Learn a mapping from inputs to outputs.
- *Experience*: Labeled examples (e.g., "This image is a cat," "This email is spam").

Unsupervised Learning (Learning by Finding Patterns)

- *Goal*: Discover hidden structures in data.
- *Experience*: Unlabeled data (e.g., "Here is a set of 10,000 news articles").

Reinforcement Learning (Learning by Trial and Error)

- *Goal*: Learn a "policy" to get the most rewards.
- *Experience*: A system of rewards and punishments (e.g., "You won the game," "You crashed the robot").



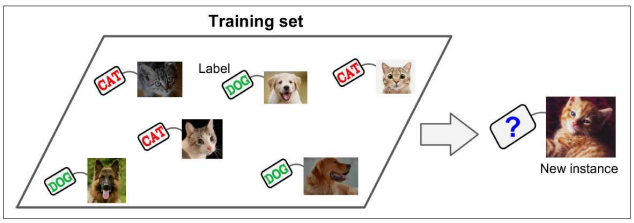
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Typical process that AI learns to classify

- **Training** (The Experience): The AI is fed thousands of objects, each with a specific label. This teaches it to “classify" the objects.

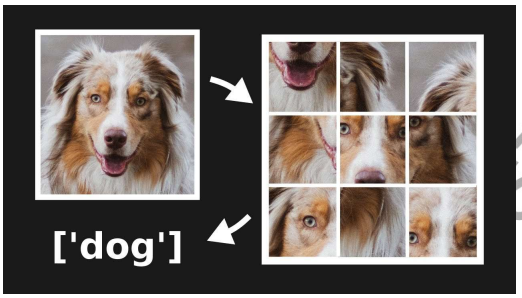


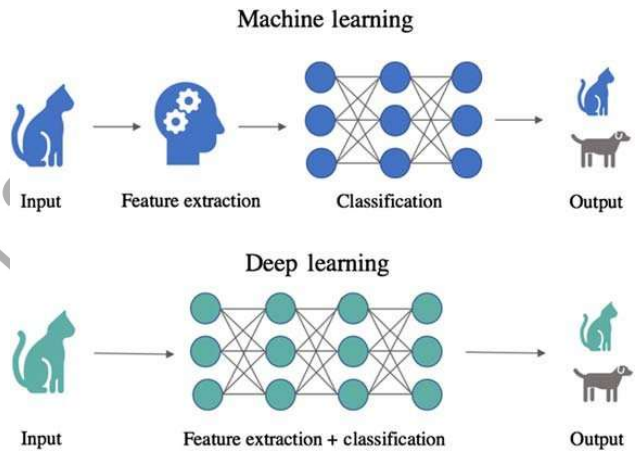
- **Learning**: The model learns the statistical patterns and features that are predictive of each label.
- **Test** (to classify): The AI is shown with a new, unseen image. It analyzes the features of this new object and applies its learned patterns to assign the correct label

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Feature extraction

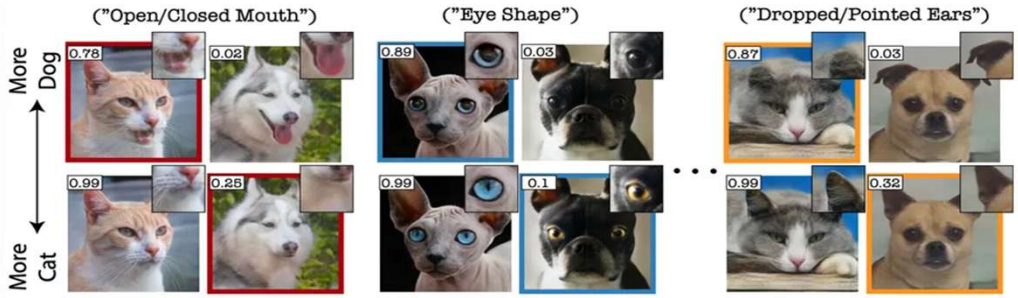




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Example of feature extraction for AI learning to classify



The feature extraction is used to classify new images that the AI model have never seen before

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Contents

- How AI learn?
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- How to learn AI?

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Open Discussions

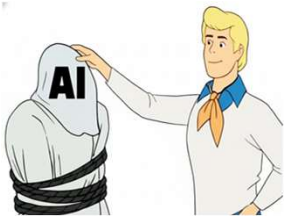
- How machine understands data to learn as intelligence?
- Is data considered as knowledge?
- Is knowledge the same as intelligence?

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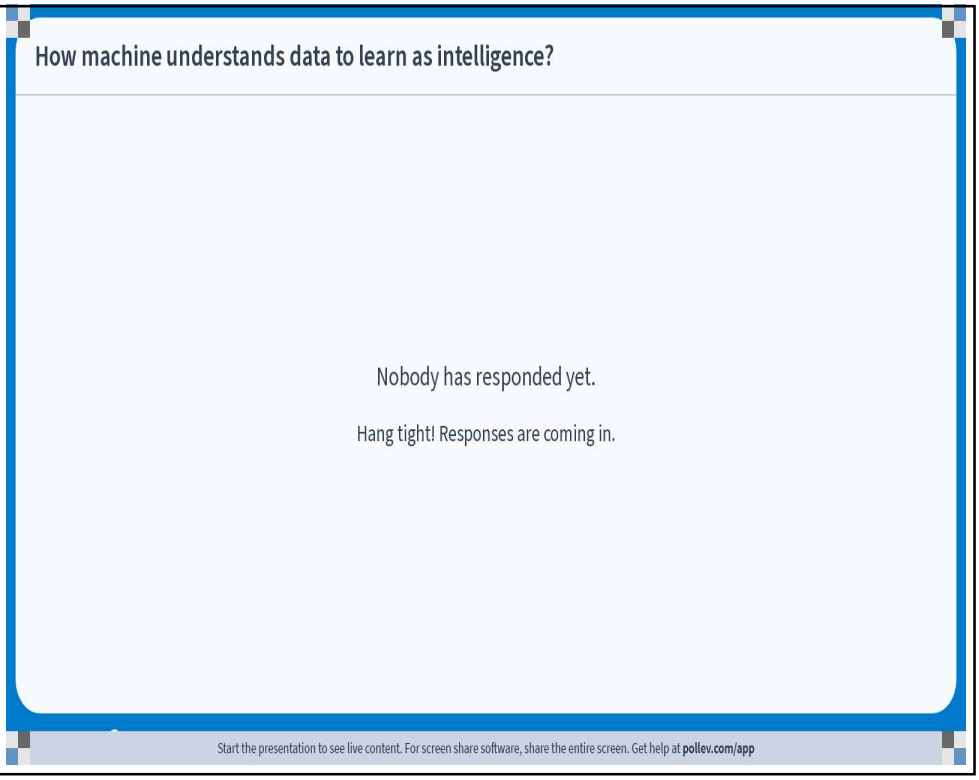
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Quiz

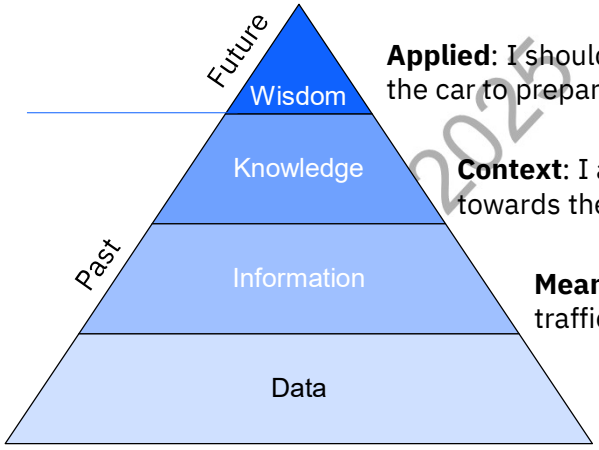


How machine understands data to learn as intelligence?



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From Data to Knowledge and Wisdom (Intelligence)



Applied: I should slow down the car to prepare the stop

Context: I am driving the car towards the red traffic light

Meaning: ahead facing traffic light has turn red

Raw: see Red

DIKW framework

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Knowledge Representation: Definition

Knowledge Representation (KR) involves encoding information in a computable form for problem-solving

Context

Originated in the 1950s-1960s with researchers such as John McCarthy and Marvin Minsky.

Importance

Facilitates retrieval of facts, relationships, and rules; essential for handling complex domains like medical diagnosis.

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Type of Knowledges in AI

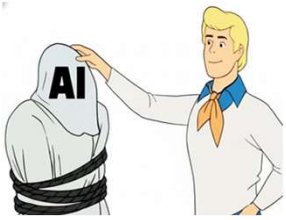
Type	Description	Example	Representation Technique
Procedural	How to preform a task	*"How to start a car"*	Rule-based (as fact)
Heuristics	Rules of experts or experience-based routes	*"if temperature is rise, stop a car"*	Rule-based with experts
Declarative	Facts or relationships	*"Hanoi is the capital of Vietnam"*	Semantic network (as fact)
Ontological	Structured meaning of concepts	*Country -> provinces -> wards -> …*	Semantic network with knowledge (knowledge graph)

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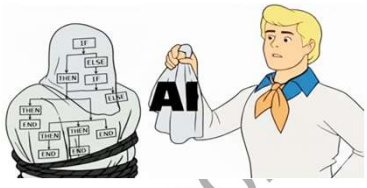
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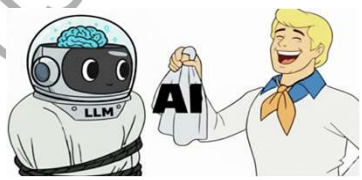
So what is inside AI?



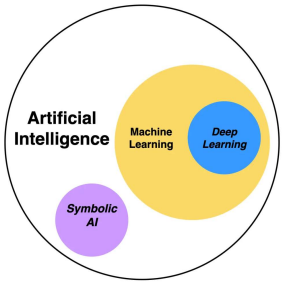
How machine understands data to learn as intelligence?



Traditional AI: Symbolic AI (rules, logic).



Modern AI: Machine learning (patterns, data).



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Key Approaches in Knowledge Representation

- Rule-Based Systems: Collections of if-then production rules.
- Frames: Structured schemas with slots for attributes and default values.
  - Example: "Bird" frame includes "wings: true," "flies: true" (overridable for exceptions like penguins).
- Semantic Networks: Graph-based structures showing relationships between concepts.
  - Example: "Dog" node linked to "Animal" via "is-a" relation.
- Knowledge Graph: also graph structure that not only shows relationships but also understands them and can infer new ones.

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Highlights for KR Approaches

Aspect	Rule-Based Systems	Frames	Semantic Networks
Structure	If-then production rules	Structured templates with slots and fillers	Graph with nodes (concepts) and labeled edges (relations)
Key Mechanism	Forward/backward chaining	Inheritance, defaults, and procedural attachments	Inheritance and associative links
Expressiveness	Conditional actions and logical implications	Object-centered with defaults and exceptions	Relations between entities (e.g., "is-a", "part-of")
Strengths	Transparent, modular, and explainable	Handles defaults and structured objects well	Intuitive visualization; supports inheritance
Limitations	Poor with uncertainty; rule explosion	Complex slot management; rigid hierarchies	Ambiguity in relation types; scalability issues
Usage	Expert systems, business rules engines	Object modeling in AI planning	Knowledge graphs (e.g., Google, Wikidata)

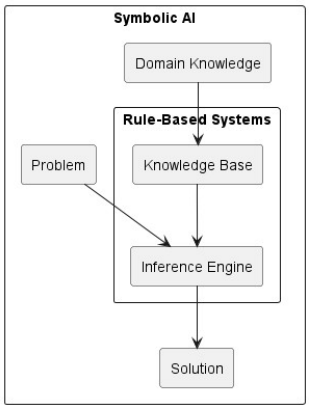
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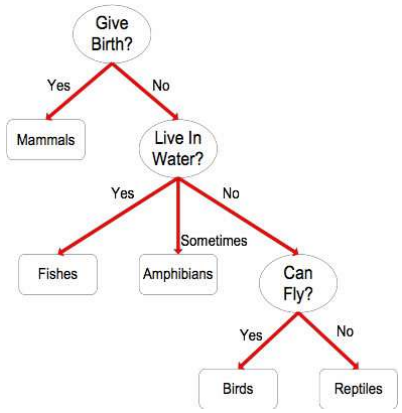
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Rule-Based Systems

- Description: Systems comprising condition-action rules.
- Mechanism: Inference engines apply rules to facts to derive conclusions.
- Advantages: Modularity and explainability; limitations in handling uncertainty.



Example: Rule-based classification (Decision tree)



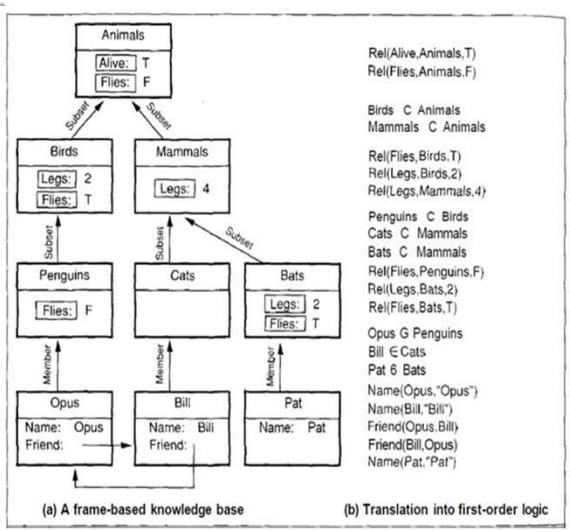
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Frames

- Description: Data structures resembling object-oriented classes with slots for properties and procedural attachments (by: Marvin Minsky 1970s).
- Mechanism: Inheritance hierarchies; default and if-needed values.
- Advantages : Common sense reasoning in expert systems.

Example: Frame-based Representation



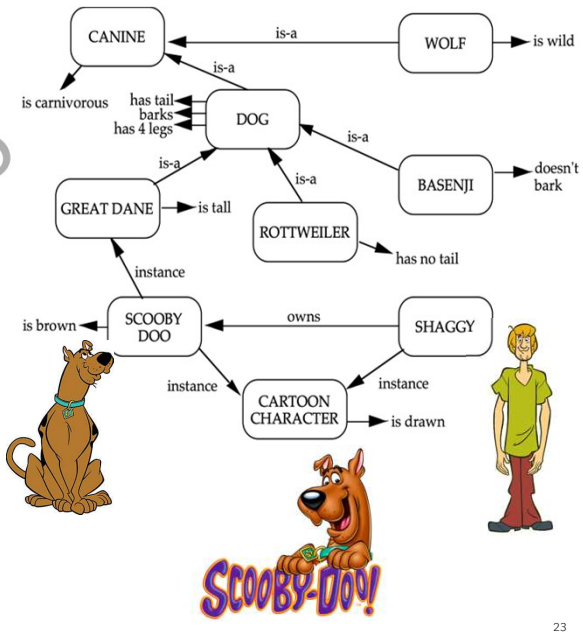
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Semantic Networks

- Description: graphs representing concepts and their interrelations.
- Mechanism: Knowledge graphs in search engines
- Advantages: Supports inheritance and inference; visualizes conceptual associations.

Example: A sample network



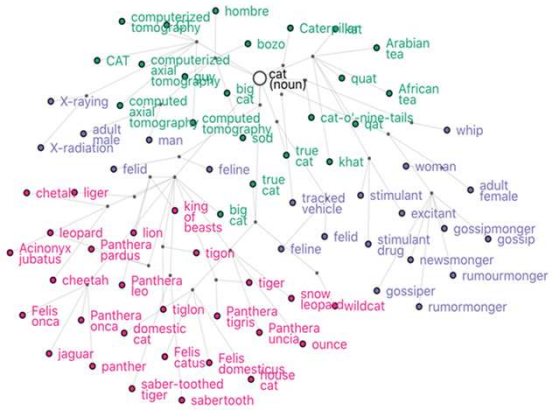
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Semantic Networks – Example with WordNet

- **WordNet** is a massive digital dictionary that connects words based on their meaning. It's a real-world example of a semantic network.
- How it Works: It treats words as nodes and their relationships as links.
  - A "car" node connects to a "vehicle" node with an is-a-kind-of link.
  - A "wheel" node connects to a "car" node with a has-part link.
- Impact: It's foundational for NLP, powering search engines and translation tools to better understand the context of human language.



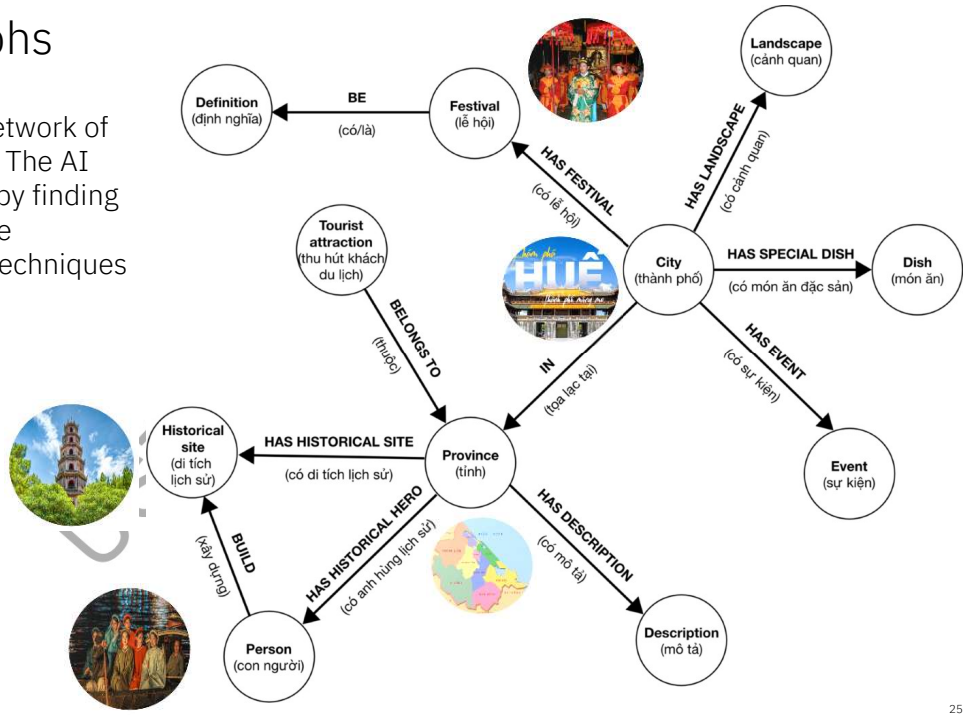
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Knowledge Graphs

Representing facts as a network of entities and relationships. The AI can "generate" new facts by finding paths in this graph, it is the searching and reasoning techniques



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Challenges in Knowledge Representation

- Ambiguity and uncertainty
- Scalability and updating knowledge
- Integrating structured and unstructured data

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How AI can learn more in new future

AI does not only learn from experience (training data), AI can also learn by reasoning and watching in near future

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From Knowledge to Reasoning

- Once knowledge is represented, AI can reason
  - AI doesn't just memorize facts (experience) — AI learns patterns in training facts, then generates likely new ones

- Fact 1: All machines need maintenance.
- Fact 2: AI server is a machine
- ➢ *New Fact: AI service needs maintenance*

Represent facts as rules
(foundation of expert systems)

- Fact: (You) → BornIn → (City X) → Located In → (Country Y)
- ➢ *New Fact: (You) have a relationship to (Country Y). Which is "LocatedIn"*

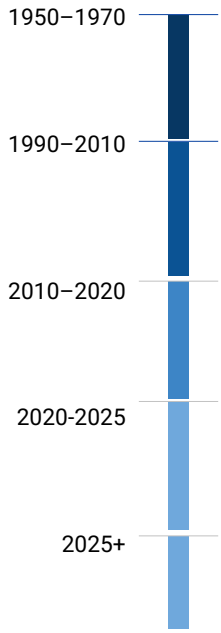
Represent facts as graph

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History of Reasoning



Symbolic (rule-based)
Systems represented knowledge explicitly with symbols and rules, using logical inference to produce interpretable conclusions but struggling with brittleness and scalability, an expert system for medical diagnosis, used hundreds of if–then rules to recommend treatments and explain its reasoning.

Probabilistic Learning
Algorithms shifted to model uncertainty and learn from data (Bayesian, Hidden Markov Models, early ML/RL), enabling robust inference from noisy sensors and statistical patterns. HMMs powered mainstream speech recognition systems by modeling temporal uncertainty in audio-to-text mapping.

Neuron Network
DNN provided rich representations that, when combined with planning or RL, solved high-dimensional perception-plus-control problems at scale. AlphaGo combined deep policy/value networks with Monte Carlo Tree Search to defeat top human Go players.

Hybrid (Neuron & Symbolic)
Large pretrained transformers enable flexible, prompt-driven multi-step reasoning and—when combined with symbolic modules, tool use, and planners—form agentic systems that decompose and execute complex tasks.

Agentic AI
GPT-style models using chain-of-thought prompting, together with tool-calling frameworks

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Symbolic AI (Rule-based)

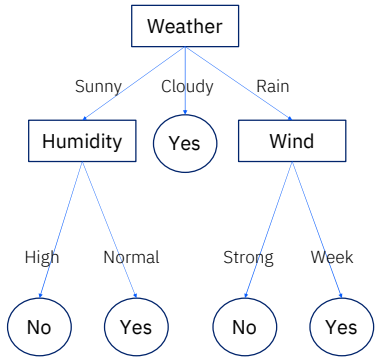
Knowledge expressed as symbols, predicates and if–then rules; formal inference with propositional or predicate logic.

Strengths: interpretable, deterministic, easy to audit in well-defined domains.

Limitations: brittle, hard to scale, and poor at handling uncertainty or raw noisy sensor data.

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Rule-based classification (Decision Tree) to play Pickleball



R1: If weather = Sunny and Humidity = High then Play = No
R2: If weather = Sunny and Humidity = Normal then Play = Yes
R3: If weather = Cloudy then Play = Yes
R4: If weather = Rain and Wind = Strong then Play = No
R5: If weather = Rain and Wind = Weak then Play = Yes

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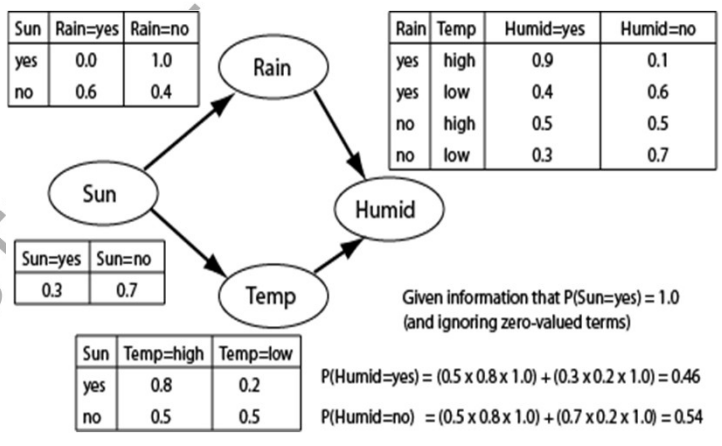
Probabilistic Reasoning

Probabilistic reasoning represents uncertainty explicitly and computes belief distributions.

- Bayesian networks, graphical models, probabilistic programming

Typical uses: robotics localization, sensor fusion, and belief/state tracking under noisy observations.

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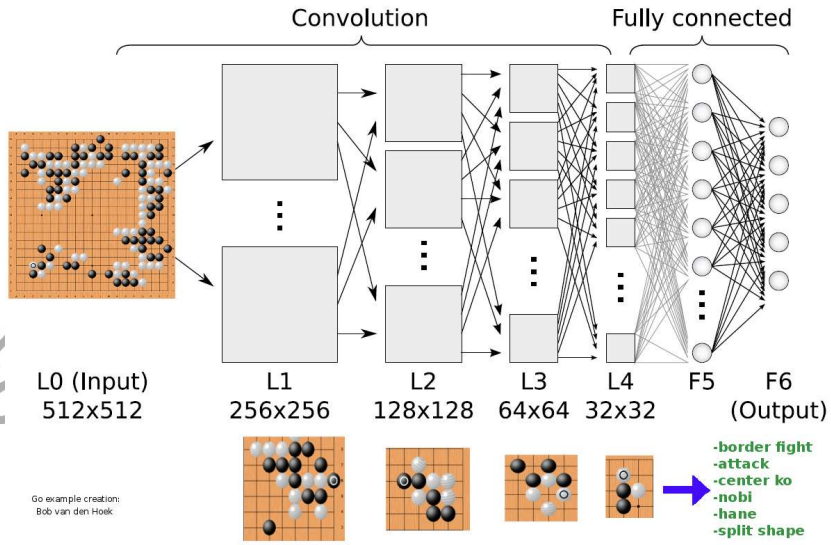
Example in weather prediction

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Neuron Network Reasoning

This image illustrates how AlphaGo's neural network reasons by first using convolutional layers to identify local patterns on the Go board, then passing those features to fully connected layers to understand more abstract, global concepts and decide on an output.

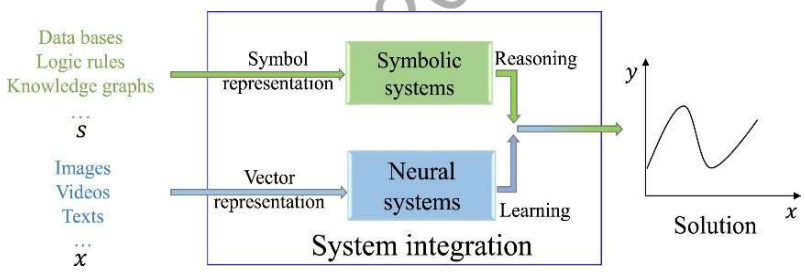


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Hybrid Approaches: Neural–Symbolic

Hybrid combination of
- neural networks' ability to learn from raw data
- symbolic methods' structured reasoning and explainability.



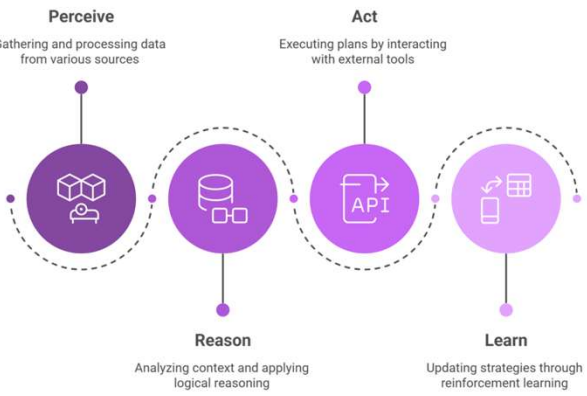
Goals: better sample efficiency, stronger generalization, and more interpretable decisions.

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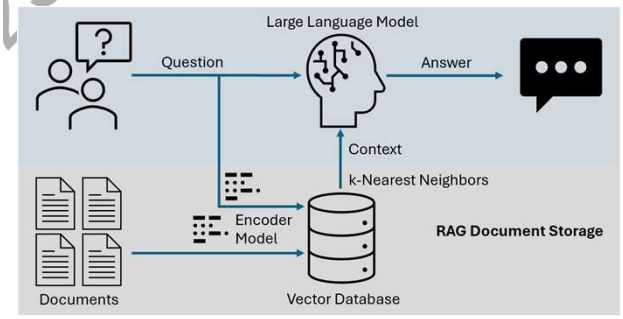
Reasoning in Agentic AI

Agentic AI Problem Solving



RAG

Retrieval-Augmented Generation



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Quiz

- What roles in IT do you want to be in the future ?



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Data & AI Jobs

Fastest growing vs. fastest declining jobs

	Top 10 fastest growing jobs	Top 10 fastest declining jobs
1.	AI and Machine Learning Specialists	Bank Tellers and Related Clerks
2.	Sustainability Specialists	Postal Service Clerks
3.	Business Intelligence Analysts	Cashiers and ticket Clerks
4.	Information Security Analysts	Data Entry Clerks
5.	Fintech Engineers	Administrative and Executive Secretaries
6.	Data Analysts and Scientists	Material-Recording and Stock-Keeping Clerks
7.	Robotics Engineers	Accounting, Bookkeeping and Payroll Clerks
8.	Electrotechnology Engineers	Legislators and Officials
9.	Agricultural Equipment Operators	Statistical, Finance and Insurance Clerks
10.	Digital Transformation Specialists	Door-To-Door Sales Workers, News and Street Vendors, and Related Workers

Source
World Economic Forum, Future of Jobs Report 2023.

Note
The jobs which survey respondents expect to grow most quickly from 2023 to 2027 as a fraction of present employment figures

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Data & AI Roles



AI Research Scientist

Invents new AI algorithms and theories to solve complex problems, often publishing their work as academic research.



AI Engineer

Turns a successful AI model into a real, working, and reliable application or feature that people can use.



Data Scientist

Applies statistical analysis and machine learning models to complex data to uncover hidden patterns and predict future outcomes.



Data Analyst

Examines historical data to identify trends, create visualizations, and answer specific business questions.



Data Engineer

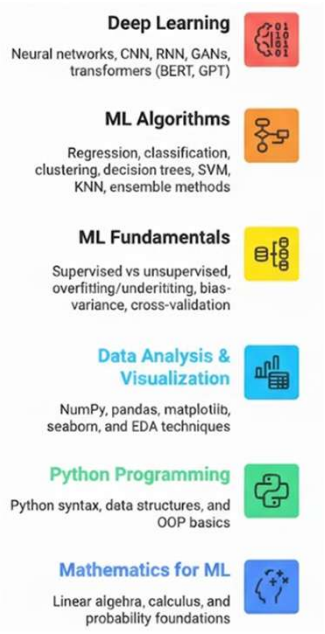
Builds and maintains the "pipelines" that collect, clean, and deliver large volumes of data.

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Data Science Roadmap



- Learn and build complex neural network architectures (like CNNs, RNNs, and Transformers) which are essential for handling very large and intricate datasets
- Learn classical machine learning algorithms like regression, classification, and clustering, to solve various predictive and descriptive problems
- Learn the differences between supervised/unsupervised learning, managing model errors like overfitting/underfitting, and effective validation techniques.
- Learn to clean, manipulate, explore (EDA), and visually represent data, which is crucial before any modeling can begin
- Mastering Python's syntax, data structures, and Object-Oriented Programming (OOP)which is the primary language for AI
- Foundational mathematical knowledge is necessary to understand how machine learning algorithms work and to analyze their performance.

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Data Science Tech Stacks/ Frameworks

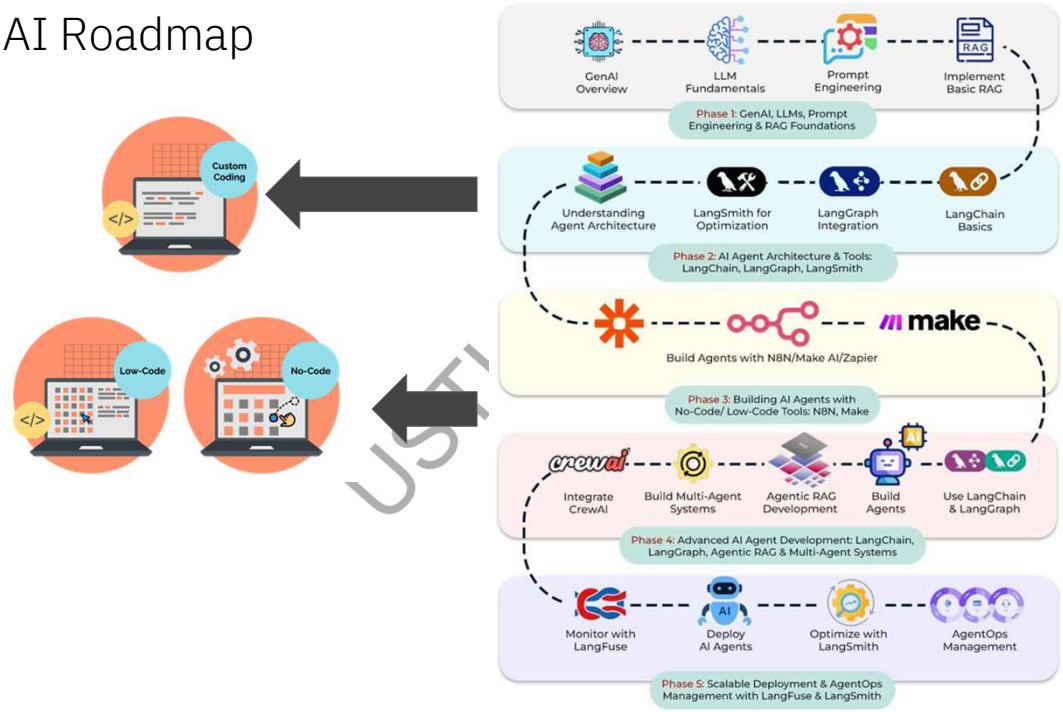


https://roadmap.sh/ai-data-scientist

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GenAI Roadmap



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Thank you!

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